1. Relational schema translation

Notes: PK – <u>bold, highlighted</u>, FK - *italic*

1. Organization relation

Organization(**PK: <u>organization_id</u>**, organization_name, org_type, phone, email, website)

This table stores information about the organizations that provide healthcare assistance programs. Each organization has a unique identifier. The table includes the organization's name, type, contact phone number, email, and website. organization_id is the primary key. organization_name should be unique and cannot be null.

2. Program relation

Program(**PK: <u>program_id</u>**, **FK:** *organization_id*, program_name, description, eligibility_criteria, benefit_type, application_deadline)

This table stores information about healthcare assistance programs. Each program is offered by exactly one organization. The table includes the program's name, description, eligibility criteria, type of benefit, and application deadline. program_id is the primary key. organization_id is a foreign key referencing Organization(organization_id). program_name and organization_id cannot be null.

3. ApplicantUser relation (previous name User)

<u>Note:</u> *User was the initial name during ER diagram phase implementation but it was discovered later that user is a reservered name in MySQL/MariaDB and that's why it was renamed to ApplicantUser*

ApplicantUser(**PK: <u>user_id</u>**, first_name, last_name, date_of_birth, email, phone, city, state)

This table stores information about users who search for and apply to healthcare assistance programs. Each user has a unique identifier. The table includes personal and contact information such as first name, last name, date of birth, email, phone number, city, and state. user_id is the primary key. email should be unique. first_name, last_name, and email cannot be null.

4. Application relation

Application(**PK: <u>application_id</u>**, **FK:** *user_id*, **FK:** *program_id*, application_date, status, notes)

This table stores applications submitted by users to healthcare assistance programs. Each application is submitted by one user and is associated with one program. The table includes the application date, status, and optional notes. application_id is the primary key. user_id is a foreign key referencing ApplicantUser(user_id), and program_id is a foreign key referencing Program(program_id). application_date and status cannot be null. ***Additionally, a user is allowed to apply to a specific program only once, which will be enforced by a UNIQUE constraint on (user_id, program_id) in SQL implementation.***

2. How the relationships were translated

From ER to relations:

Organization offers Program (1:N)

This is implemented by putting organization_id as a foreign key inside **Program**.

ApplicantUser submits Application (1:N)

This is implemented by putting user_id as a foreign key inside **Application**.

Program receives Application (1:N)

This is implemented by putting program_id as a foreign key inside **Application**.

So the foreign keys are:

- Program.organization_id -> Organization.organization_id
- Application.user_id -> ApplicantUser.user_id
- Application.program_id -> Program.program_id